

Demonstration of Flashpen: A high-fidelity and high-precision multi-surface pen for Virtual Reality

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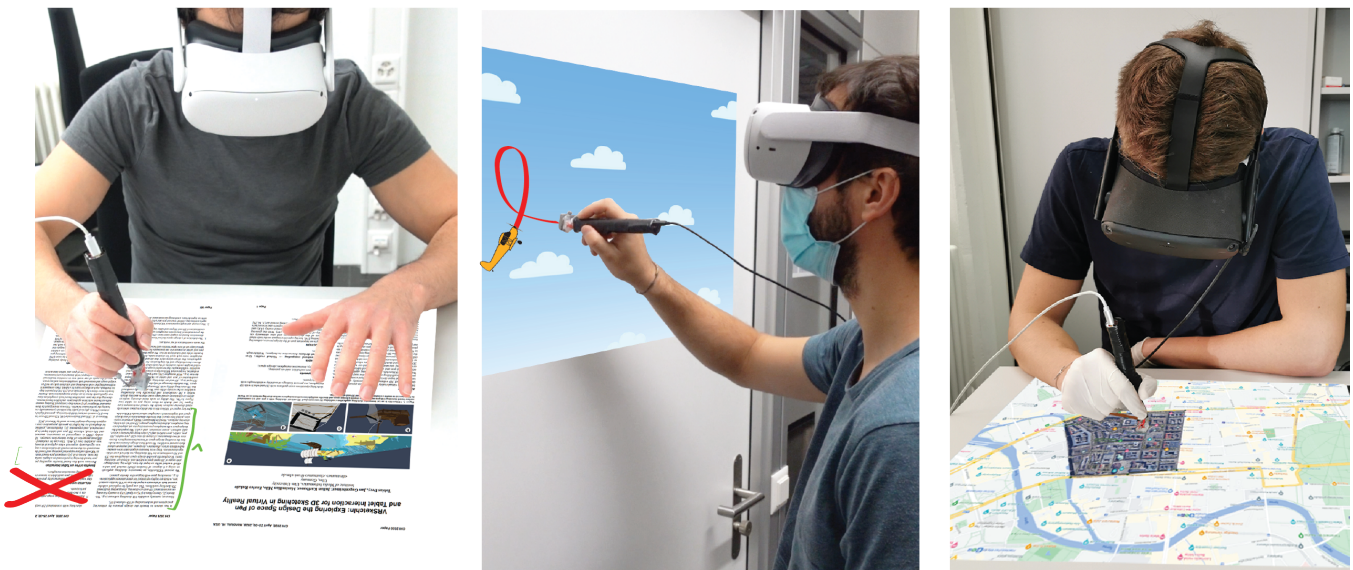


Figure 1: FlashPen can be used in a virtual reality environment with the user situated at a horizontal or vertical surface (virtual content superimposed in this figure to demonstrate system usage). This enables several VR pen interactions including document annotation (left), guiding animated objects (middle), and using a multi-layered map interface with a magic lens metaphor (right).

ABSTRACT

Pen computing has become popular with tablet and wall-screen computers for digital precision tasks such as writing, annotating, and drawing. Digital pens have been made possible by the developments in input sensing technologies integrated into such screens.

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Virtual Reality systems, however, largely detect input using cameras, whose update rates are insufficient for capturing pen input with the necessary fidelity. In this demonstration, we showcase a digital pen for VR that accurately digitizes writing and drawing, including small and quick turns. Our prototype Flashpen repurposes an optical flow sensor from gaming mice, which digitizes minute motions at over 8, kHz when dragged across a surface.

We demonstrate several use-cases for Flashpen during interaction in VR, including sketching, selecting, annotating, and writing.

KEYWORDS

Human-centered computing, Human computer interaction (HCI), Interaction devices, Graphics input devices

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1 INTRODUCTION

Pen interaction is becoming a commodity input modality on interactive devices [2, 14], and has been shown to help externalizing thoughts [11] via annotations and note-taking and enhance visual thinking [16]. With the growing availability of Virtual Reality applications and platforms, designers and researchers have started to explore their capability of supporting pen input for sketching [7, 8], to support gestures input [12], or to support mid-air techniques [3]. The challenge of supporting high-quality pen input in VR is that pen input can be minute, requiring a precision mostly between 0.1 and 1 mm [4, 15] due to the size of the tip to digitize strokes with high fidelity. The accuracy is thereby limited by the tracking system, i.e., the resolution and framerate of the cameras used for registering input locations. The former is sufficient for sketching under visual control and has been shown to support an accuracy between 1 mm to around 5 mm in a dynamic configuration for the controller [5, 10]. However, it is the camera's framerates, mostly around 65Hz for most headsets, that are most limiting when reconstructing input strokes, which are the crucial element of inking on surfaces [6] and contribute to users' lack of perception that such writing is similar to analog writing on paper.

In this paper, we propose a complementary approach to further approach analog pen input qualities in VR systems. Inspired by the inside-out tracking approach taken by the latest headsets, we present Flashpen, a prototype stylus with an optical flow sensor at its tip. For this purpose, we reappropriate a flow sensor found in high-end gaming mice that operates at a framerate of 8 kHz, thus digitizing pen strokes at a rate that is 40 times higher than that of state-of-the-art stylus input digitizers (e.g., Wacom [17], Microsoft Surface Pro [14]). Our implementation fuses Flashpens high-frequency relative motions with the absolute locations from the VR tracking system, bringing absolute pen input to VR systems that digitize handwriting and sketching at near-analog fidelity. We demonstrate Flashpen in a series of VR applications and highlight the potential of our approach. In our evaluation of the prototype in a 10-participant user study, we compared its performance to that of a WACOM pen stylus, finding a fidelity that is comparable to handwriting on paper for our prototype while matching WACOM pen's precision.

2 IMPLEMENTATION

Our prototype is shown in Figure 2. FLASHPEN is designed as a convention pen or stylus, held with two or three fingers during interaction. All hardware components are embedded in the shell of the pen, which is made from a 30 mm plastic tube, wrapped in black electrical tape and encases all components except the tip. Because users hold pens with various poses while writing with a regular pen, FLASHPEN integrates a light plastic joint (Lego Mindstorm hinge, Figure 2.4). FLASHPEN can thus be used by left-handed and right-handed users and supports various grips for holding.

To track pen motions, FLASHPEN embeds a Pixart PMW3360 flow sensor at the tip of the pen (Figure 2.2) whose lens and cover (Figure 2.3) we sanded down to be as small as possible. An InvenSense

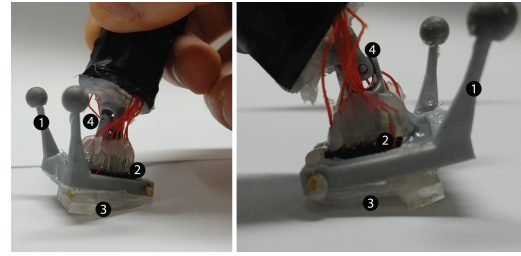


Figure 2: FLASHPEN prototype: (left) full pen device, (right) zoomed in on the tip of the pen with (1) markers for tracking, (2) optical flow sensor, (3) sensor lens, and (4) a plastic hinge to accommodate writing at various angles.

MPU6500 inertial sensor mounted inside the shell of the pen captures rotations and accelerations. All parts connect to an Expressif ESP32 microcontroller, which is tethered to the VR headset for power purposes and to minimize the latency of input sensing. In addition, reflective markers are mounted around the pen tip (Figure 2.1) for tracking absolute positions using two surrounding cameras (OptiTrack Flex13). Overall, FLASHPEN weighs approximately 200 g.

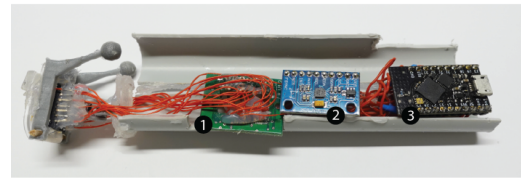


Figure 3: FLASHPEN's electronic components: (1) PMW3360 breakout board (optical flow sensor), (2) MPU6500 breakout board (IMU sensor), (3) ESP32 prototyping board (TinyPico)

The flow sensor inside FLASHPEN resolves relative motions at 8,000 fps, encoded in fractions of a millimeter. We configured the flow sensor to operate in burst mode, such that FLASHPEN's microcontroller can report all relative motion events to the VR system at the full framerate and therefore highest fidelity of digitization. To complete FLASHPEN's position tracking, we complement the high-speed relative motion sensing on the pen with absolute positions from a camera system that records the reflections from FLASHPEN's markers. While this tracking could be implemented based on the cameras in the VR headset, we opted for a small but stationary tracking system instead. This precludes additional errors that may be caused by head motions and that would thus impact pen tracking accuracy and manifest in lower-fidelity digital writing.

Note that while FLASHPEN's two tracking components are complementary, we only ever use one modality at a time as the dominant source of tracking information. Starting with mid-air motions, our system tracks FLASHPEN using the pair of cameras. FLASHPEN's flow sensor detects the presence and contrast of reflections, from which our prototype infers that it has been set down on a surface. Setting down the pen is an event that also shows up in the continuous signal of the inertial sensor. Once this state has been detected, tracking shifts to the pen into a relative mode, which the pen resolves

much faster and more precisely than the camera system, streaming positions to the headset.

3 DEMONSTRATIONS AND USE CASES

In all our applications, we represent FLASHPEN as a virtual pen to maintain functional consistency between the physical world and the virtual world. In order to establish a frame of reference, our system can also display the left hand in VR using the Oculus Quest hand pose tracking.

MAP APPLICATION - In this example, a map view is displayed in VR where the pen operates as a cutting tool. Any shape can be drawn by placing the pen down and cutting out the part of the map that should be replaced by a satellite view. Once a point has been drawn, users can adjust it by placing the pen tip on it and moving it around to precisely adjust its position. Once the polygon shape has been defined, the pen acts as a precise manipulation tool to move the shape around like a magic lens, revealing the layer below.



Figure 4: Magic lens interaction with a two-layered map view. The user drags a lens that renders the satellite layer of a map.

DRAWING APPLICATION - We also demonstrate FLASHPEN's high fidelity and precision in a painting app where FLASHPEN allows users to precisely fill shapes and interact with widgets, such as sliders. The sliders, as shown in Figure 5 (bottom right), can also be used to control the RGB color as well as the size of the stroke to draw. Users can position the pen inside the slider area and touch the surface to select an absolute value. The user can then also drag the slider by precisely dragging the pen until the exact desired value is reached. Once the user selected a color, he or she can start drawing or filling shapes that are already displayed. This very precise on-surface manipulation would not be possible using a regular VR pointer and the tracking in current headsets or even 6-DOF tracking systems. Our computation of pen-down locations using external cameras is a first step, but FLASHPEN's true capabilities come to fruition during relative motions on the surface.



Figure 5: The user can select a color and thickness to colorize a shape. With fine-grained pen movements, the user can accurately fill small details of the shape with color.

REVIEWING DOCUMENTS - By combining FLASHPEN's relative high-fidelity tracking with our absolute positioning using the Opti-track cameras, we register absolute positions on the surface. Here, users can operate FLASHPEN to review a document by highlighting important parts, jotting down comments, and emphasizing paragraphs of interest. Using a swipe gesture with the non-dominant hand, users can switch between pages to review several pages while immersed in a VR environment, which we derive from the headset's hand pose tracking. Related to this application, we see an opportunity for FLASHPEN as the basis of an exploration of broader ranges of scenarios and interaction techniques that could involve pen+touch [9] in VR.



Figure 6: The user annotates a document. Pages are turned with hand gestures (non-dominant hand).

MOTION PATH ANIMATION - As shown in Figure 7, we use the high sample rate of FLASHPEN in this application to design motion experiences where the user draws strokes using varying input speeds to guide the animation of a plane. Such motion paths, thus, simultaneously encode spatial as well as temporal information, for example by drawing slowly while going in a straight line and then accelerating while drawing a looping. When the user lifts the pen, the yellow plane shown in Figure 7 (bottom right) will follow the exact same path with the same velocity. In addition, this application demonstrates FLASHPEN's suitability to be usable on several surfaces, horizontal and vertical alike.

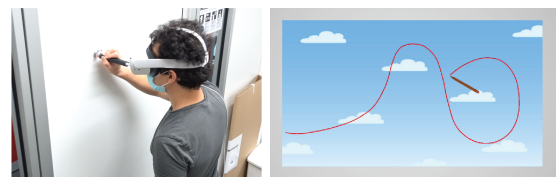


Figure 7: An animated plane is following the user-defined spatio-temporal curve, i.e., with varying velocities to exactly replicate the defined motion.

4 CONCLUSION

Our demonstration showcased high-fidelity pen input in Virtual Reality. We take a complementary approach of camera-based absolute tracking and pen-based high-speed inside-out tracking for fast-occurring relative motions. In a series of applications, we show the potential of inking in VR, whose quality and fidelity is uniquely enabled by our prototype FLASHPEN.

FLASHPEN is a novel input device that allows users to precisely sketch and draw at a high sample rate, uniquely enabling the almost perfect reconstruction of the analog pen motions. The results of the

study we have run, both pertaining to participant accuracy, perceptual responses, and feedback, strongly suggest that our prototype is a promising tool for precise writing and 2D manipulation.

However, when writing, users involve numerous muscles and skills [13], which essentially render the pen an extension of the user's hand. Longitudinal studies will be needed to fully understand the impact of our prototype on the more general writing experience, as well as on users' comprehension.

Pen input in VR, as we have shown, is also an opportunity to design novel interaction techniques that go beyond the traditional controllers and hand interactions usually found in current VR applications. We think that high-fidelity pen interaction is a promising tool for precise input, which could be exploited for new interaction mechanisms such as pen+touch interaction. With this paper, we show the first points of the more general design space that FLASHPEN opens up, and future work will be needed to investigate this rich space thoroughly.

In our fused tracking system, we have used Optitrack cameras as a stationary sensing technology to detect pen-down and lift-off gestures to enable absolute positioning of the pen. However, for some users, fast movements above the surface when lifting the pen, and lifting gestures might not be as accurate as an analog pen would be. To increase precision, other absolute tracking technologies, such as the Anoto pen [1], could be used in a complementary manner. Also, as mentioned earlier, the ergonomic factor of our pen, which is currently designed as a cylinder, might affect the way users write. As they are often used to writing with more fusiform devices, such as analog pens, this form factor has to be taken into consideration to match what currently exists on the market and to establish a user experience that is closer to what they have experience with.

Currently, FLASHPEN performs well on most surface types. In our empirical tests, we tried numerous surfaces, such as wood, paper, or plastic, which all resulted in similar precisions—unsurprisingly, as mouse sensors are designed to function properly on such surfaces. However, much like with mouse sensors, transparent surfaces such as windows or glass can prove challenging and the tracking might look jittery as the infrared light is not reflected properly. One alternative could be the use of blue light sensors in the future to accommodate more surface types. Another factor to consider in the design of future iterations of FLASHPEN is the lens, which is

situated at the pen tip. Over the course of use, the lens may wear off depending on surface materials or get stuck on features such as surface imperfections or creases on clothing.

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